

DESIGN DESIGN

Product: DeepMindQ Enterprise Intelligence OS
Phase: 1B — Executive Intelligence Experience
Status: Design Review

A comprehensive design document defining the transformation of the DeepMindQ Command Center into a primary executive intelligence surface. Covers information architecture, intelligence hierarchy, visual design, and acceptance criteria for the VP Sales/CRO first-5-minute journey.

Command Center as the Executive Intelligence Experience

Table of Contents

Chapter 1: Executive Summary	3
1.1 Phase 1A Achievements at a Glance	3
Chapter 2: Current State After Phase 1A	4
2.1 Command Center Architecture	4
2.2 Intelligence Pipeline Integration Map	4
2.3 What Works Today	5
2.4 What Remains Incomplete	5
Chapter 3: User Journey Problems Remaining	6
3.1 The 5-Minute Test Failure Points	6
3.2 Navigation Overload Analysis	7
3.3 Context Loss Between Screens	8
Chapter 4: Target Experience	8
4.1 The VP Sales First-5-Minute Journey	8
4.2 The 5 Questions, Answered in Under 5 Minutes	9
Chapter 5: Information Architecture	10
5.1 Primary Zones	10
5.2 Intelligence Hierarchy	11
5.3 Cards, Panels, and Feed Behavior	11
5.4 Navigation Reduction	12
5.5 Context Preservation	13
Chapter 6: Visual and Emotional Design	13
6.1 Why This Experience Feels Fundamentally Different	13
6.2 Emotional Design Principles	15
6.3 Intelligence Advantage Feel	15
Chapter 7: Technical Implementation Strategy	16
7.1 Component Architecture	16

7.2 Data Flow Architecture	17
7.3 Intelligence Pipeline Extensions	17
Chapter 8: Acceptance Criteria	18
8.1 Functional Acceptance Criteria	18
8.2 Technical Acceptance Criteria	19
8.3 Known Limitations for Phase 1B	20

Chapter 1: Executive Summary

Phase 1A of the DeepMindQ Product Experience Transformation established the intelligence design system foundation, created the core intelligence OS component library, connected the Command Center to the real intelligence pipeline, and proved that the product can deliver intelligence-driven experiences rather than static dashboard views. The transformation was verified through five technical corrections, each addressing a critical gap between visual representation and real engine connectivity.

The fundamental architectural shift achieved in Phase 1A moved DeepMindQ from a paradigm of "Data, Dashboard, User interpretation" to "Intelligence, Reasoning, Evidence, Confidence, Recommendation, Human action." This is not merely a visual redesign; it represents a structural reorientation of how the product communicates value to its users. Every intelligence narrative now carries real confidence scores computed from multi-factor formulas, traceable evidence chains sourced from actual signals and documents, and actionable recommendations generated by the AI action engine.

Phase 1B now takes this foundation and elevates it to the executive experience level. While Phase 1A proved the intelligence pipeline works within the Command Center component, Phase 1B must prove that a VP Sales or CRO can open DeepMindQ and, within five minutes, understand what needs attention, why it matters, what evidence supports the assessment, and what action to take. This is the standard against which the product will be evaluated, and it requires a fundamentally different approach to information architecture, visual hierarchy, and contextual intelligence delivery.

1.1 Phase 1A Achievements at a Glance

Component/Service	Phase 1A Achievement
IntelligenceNarrative	Connected to real engine outputs via NarrativeService
ConfidenceIndicator	Real 4-factor calculation (Signal 30%, Evidence 30%, Capability 25%, Data 15%)
EvidenceChain	Traceable evidence from GroundingEngine with source citations
ActionCTA	Recommendation-driven actions from ActionEngine
Command Center	Consumes full intelligence pipeline via useIntelligenceNarratives hook
Test Coverage	40+ new tests (confidence formula, API shape, VP Sales 5-question validation)
Components Delivered	10 intelligence OS components with unified design tokens
Services Connected	6 engines (Grounding, Synthesis, Scoring, Action, Retrieval, Model Router)

Phase 1B Mandate: Phase 1A proved the intelligence foundation. Phase 1B must now prove the experience at executive level. A VP Sales/CRO opens DeepMindQ and within 5 minutes understands: What needs attention today? Which accounts have movement? Why does AI believe this? What action should happen next? How does DeepMindQ continuously learn?

Chapter 2: Current State After Phase 1A

2.1 Command Center Architecture

The Command Center, implemented in `src/components/intelligence-os/command-center.tsx`, is the primary intelligence consumption surface in DeepMindQ. Following Phase 1A, it integrates with the real intelligence pipeline through the `useIntelligenceNarratives` hook, which fetches narrative data from the `/api/intelligence/narratives` API endpoint. This API, in turn, invokes the `IntelligenceNarrativeService`, which orchestrates the full engine pipeline: Signal extraction, `GroundingEngine` evidence collection, multi-factor confidence computation, and `ActionEngine` recommendation generation.

The current Command Center renders intelligence narratives through the `IntelligenceNarrative` component with progressive disclosure layers (L1 Decision, L2 Reasoning, L3 Evidence, L4 Explore). Each narrative displays a confidence ring computed from four weighted factors, an evidence chain with source citations and relevance scores, and action recommendations with priority levels. The aggregated intelligence health bar provides a system-wide confidence overview. The "Refresh Intelligence" button triggers a manual pipeline refetch, and loading and error states are handled with appropriate visual indicators.

2.2 Intelligence Pipeline Integration Map

The following table documents the complete data flow from intelligence engines to the Command Center UI surface. Each layer in this pipeline has been verified to carry real data (not static templates) following the Phase 1A correction cycle.

Layer	Output Type	Description
Signal Extraction	CompanySignal, AIInsight, NewsSignal	Raw signals from DB, web monitors, external sources
GroundingEngine	EvidenceChain, EvidenceGap[]	Multi-source evidence collection with gap detection
Confidence Computation	NarrativeConfidence (0-100)	4-factor formula: Signal 30% + Evidence 30% + Capability 25% + Data 15%

Layer	Output Type	Description
SynthesisEngine	Brief with [En] citations	LLM-powered narrative generation with hallucination detection
ActionEngine	ActionResult with priority	6 action types: next_best_action, sales_motion, account_strategy, etc.
IntelligenceNarrative Service	IntelligenceNarrativeData[]	Bridge layer composing engines into UI-ready data
/api/intelligence/narratives	JSON with success/meta/data	3 modes: command center, signal drill-down, confidence detail
useIntelligenceNarratives	narratives[], isLoading, error	Client hook with polling, drill-down, refetch capabilities
CommandCenter	Rendered intelligence narratives	Progressive disclosure, confidence rings, evidence chains, actions

2.3 What Works Today

The Command Center successfully demonstrates the intelligence-driven paradigm when a user navigates to it. Intelligence narratives display real signals detected from monitored companies, confidence scores reflect actual evidence quality and signal strength, evidence chains trace back to specific source documents and news articles, and action recommendations are generated by the AI action engine based on grounded reasoning. The progressive disclosure pattern allows users to quickly scan L1 decision-level summaries and drill into L3 evidence and L4 exploration when they need deeper understanding.

The design token system provides a unified visual language across all 10 intelligence OS components. Confidence indicators render consistently in ring, bar, badge, and score modes. The intelligence panel provides contextual intelligence for any entity (company, contact, opportunity) through a slide-over surface that composes narratives and evidence chains. The action CTA component ensures every intelligence narrative terminates with a clear, priority-weighted next step.

2.4 What Remains Incomplete

Despite the Phase 1A achievements, several critical gaps remain that prevent the Command Center from delivering the executive-level experience required for Phase 1B closure. First, the intelligence pipeline is only connected to the Command Center. The other 76+ screens in the application still rely on static data, template-driven components, or direct API calls that bypass the intelligence narrative service. This means the intelligence-first experience is isolated to a single screen rather than permeating the entire product.

Second, the Command Center currently displays intelligence as a flat list of narratives without temporal prioritization, account grouping, or severity-based attention routing. A VP Sales arriving in the morning sees all narratives ranked by confidence, but cannot quickly distinguish "what changed since yesterday" from "what has been true for weeks." Third, the navigation architecture still presents 77+ screens through a complex sidebar, forcing executive users to choose between "Command Center," "AI Command Center," "Revenue Intelligence," "Signal Intelligence," "Account Intelligence," and other similarly named options that create cognitive overload rather than clarity.

Gap	Description
Pipeline Isolation	Intelligence pipeline connected only to Command Center; 76+ screens still static
Temporal Blindness	No "what changed since yesterday" differentiation; flat narrative list
Navigation Overload	77+ screens with 10+ "intelligence" labeled options in sidebar
Context Loss	No cross-screen context preservation; drilling into an account loses Command Center state
Feedback Gap	Accept/dismiss actions not tracked; no learning loop from user decisions
Real-Time Limitation	Polling-based only; no WebSocket for live intelligence updates
Executive Framing	No morning briefing, no daily digest, no attention-priority routing

Chapter 3: User Journey Problems Remaining

3.1 The 5-Minute Test Failure Points

The definitive evaluation criterion for Phase 1B is whether a VP Sales or CRO can open DeepMindQ and, within five minutes, answer five critical questions: What needs attention today? Which accounts have movement? Why does AI believe this? What action should happen next? How does DeepMindQ continuously learn? Currently, the product fails this test at multiple points, and understanding each failure is essential to designing the Phase 1B solution.

Failure Point 1: No Attention Routing. When the VP Sales opens DeepMindQ, the Command Center displays a list of intelligence narratives, but there is no mechanism to highlight which items are new since the last session, which items have escalated in severity, or which items are most time-sensitive. The user must scan every narrative to determine what deserves immediate attention, which defeats the purpose of an intelligence command system. A true intelligence experience should proactively surface the top 3 items that demand immediate executive attention, with a clear explanation of why each was selected.

Failure Point 2: No Account Movement Clarity. While the Command Center shows signals per company, it does not aggregate movement across the portfolio. The VP Sales cannot quickly see "3 accounts had positive movement, 1 had negative movement, and 2 are approaching decision windows." This portfolio-level view is essential for executive triage and should be the first thing visible after the attention-routed items.

Failure Point 3: Evidence Requires Too Many Clicks. To understand why AI believes something, the user must expand a narrative (L1 to L2), then expand the evidence chain (L2 to L3), then potentially click into a source link (L3 to L4). This three-click drill-down is too deep for an executive who needs to quickly validate AI confidence. The evidence should be visible at L1 or L2 for high-severity items, with progressive disclosure reserved for lower-priority intelligence.

Failure Point 4: Actions Lack Contextual Urgency. Action recommendations are displayed with priority labels (critical, high, medium, low) but without temporal context. A "critical" action from three weeks ago is not the same as a "critical" action from today. The user cannot distinguish between "act now" and "act this week" without manually checking timestamps.

Failure Point 5: Learning Loop Is Invisible. The fifth question, "How does DeepMindQ continuously learn?" is currently unanswerable from the UI. The system has learning mechanisms (feedback collection, signal recalibration, model fine-tuning) but none of this is visible to the user. The executive cannot see how their actions, accepts, dismisses, or feedback contribute to system improvement. This transparency gap undermines trust in the intelligence platform.

3.2 Navigation Overload Analysis

The current navigation architecture presents a fundamental challenge to the executive experience. The sidebar contains 77+ screen options organized into categories that overlap significantly. Multiple screens serve similar purposes with unclear differentiation: "Command Center" vs. "AI Command Center" vs. "Revenue Intelligence" vs. "Signal Intelligence" vs. "Account Intelligence" vs. "Intelligence Analytics" vs. "Intelligence Reasoning" vs. "Intelligence Health." For an executive user, this proliferation of "intelligence" screens creates confusion rather than clarity.

Category	Count	Screens
Intelligence Screens	12+	Command Center, AI Command Center, Revenue Intelligence, Signal Intelligence, Account Intelligence, Intelligence Analytics, Intelligence Reasoning, Intelligence Health, Intelligence Sources, Intelligence Knowledge, Internal Intelligence, Intelligence Inbox
Company Screens	6+	Companies, Company Detail, Company Mind Map, Company Profile, Account Intelligence, Account Ranking
Sales Execution	8+	Pipeline, Opportunities, Contacts, Sequences, Drafts, Replies, Sales Execution, Deal Coaching

Category	Count	Screens
Operations	8+	Settings, Audit, Data Health, Duplicates, Import, Analytics, Reports, Tasks
AI Features	10+	Research Agent, Conversation Studio, Conversation Planner, Email Generation, AI Strategy, AI Health, AI Usage Dashboard, AI Reasoning, Knowledge Library, Prompt Templates
Total	77+	Cognitive overload for executive users

3.3 Context Loss Between Screens

When a user drills into an intelligence narrative on the Command Center and navigates to a company detail screen, all Command Center context is lost. The company detail screen loads its own data independently, without awareness of the intelligence narrative that led the user there. This forces the user to mentally reconstruct the context: "I came here because the AI detected a hiring signal at this company, and the evidence was from a LinkedIn post two days ago." The intelligence panel partially addresses this by providing a slide-over contextual view, but it is not automatically triggered when navigating from an intelligence narrative.

This context loss is particularly damaging for the executive journey because it breaks the "intelligence narrative to action" flow. The VP Sales sees an intelligence item, decides to investigate further, navigates to the company, and then must re-find the intelligence that prompted the investigation. A true intelligence command system would preserve this context seamlessly, carrying the narrative thread across screen transitions and providing breadcrumb-level traceability back to the originating intelligence.

Chapter 4: Target Experience

4.1 The VP Sales First-5-Minute Journey

The target experience for Phase 1B is defined by the VP Sales first-5-minute journey. This journey is not a linear sequence of clicks; it is a layered intelligence consumption experience where each layer answers one of the five critical questions with increasing depth. The design must ensure that a VP Sales or CRO, opening DeepMindQ for the first time each morning, can achieve full situational awareness within 300 seconds without training, without confusion, and without feeling like they are using a "dashboard."

Seconds 0-30: Attention Routing Layer. The Command Center opens with a brief, AI-generated morning intelligence summary at the top. This is not a generic greeting; it is a dynamically composed briefing that states: "3 accounts require your attention today. Acme Corp escalated to critical after a CTO departure signal. Meridian Technologies entered a buying window based on 4 new signals. DataStream

Inc. has 2 actions approaching deadline." The summary uses natural language generated by the SynthesisEngine, grounded in actual signal data, and directly answers "What needs attention today?"

Seconds 30-90: Portfolio Movement Layer. Below the attention summary, a portfolio intelligence view shows account movement across the entire monitored portfolio. This view uses a visual grid or heat-map style layout where each monitored account is represented as a tile. Tiles are color-coded by movement direction (positive, negative, neutral), sized by revenue impact, and annotated with the top signal type. This answers "Which accounts have movement?" in a single visual scan without requiring the user to read individual narratives.

Seconds 90-180: Evidence and Reasoning Layer. When the VP Sales clicks on an account tile or attention item, the Intelligence Panel opens with the full narrative for that account. The evidence chain is visible at the first expansion level (not buried three clicks deep). The confidence breakdown shows the four contributing factors with human-readable explanations: "Signal strength: High (3 new signals in 48 hours). Evidence quality: Medium (2 confirmed sources, 1 unverified). Capability match: Strong (alignment score 87). Data freshness: Good (last updated 6 hours ago)." This answers "Why does AI believe this?" with transparent, auditable reasoning.

Seconds 180-240: Action Layer. Each intelligence narrative terminates with one or more action recommendations that include temporal urgency: "Schedule executive call this week (window closing in 14 days)" or "Send technical whitepaper within 48 hours (proposed by AI based on buying stage analysis)." Actions are prioritized not just by severity but by time-sensitivity, helping the VP Sales distinguish between "act now" and "act soon." This answers "What action should happen next?" with actionable, time-bound recommendations.

Seconds 240-300: Learning Transparency Layer. A subtle but persistent "Intelligence Health" indicator shows how the system is learning from user interactions. When the user accepts or dismisses an intelligence item, a brief micro-interaction confirms the feedback was recorded. A collapsible "Why did AI show me this?" panel explains the ranking rationale. This answers "How does DeepMindQ continuously learn?" by making the feedback loop visible without being intrusive. The key insight is that the learning transparency should not require a separate screen; it should be embedded in every interaction.

4.2 The 5 Questions, Answered in Under 5 Minutes

Question	Time Target	Phase 1B Delivery Mechanism
What needs attention today?	0-30 sec	Morning Intelligence Summary (AI-generated, signal-grounded)
Which accounts have movement?	30-90 sec	Portfolio Intelligence Grid (visual heat-map, color-coded movement)

Question	Time Target	Phase 1B Delivery Mechanism
Why does AI believe this?	90-180 sec	Evidence Chain at L1 expansion with 4-factor confidence breakdown
What action should happen next?	180-240 sec	Time-bound action recommendations with urgency indicators
How does DeepMindQ learn?	240-300 sec	Embedded feedback indicators on every intelligence interaction

Chapter 5: Information Architecture

5.1 Primary Zones

The Phase 1B Command Center information architecture restructures the current flat narrative list into five hierarchical zones, each serving a distinct cognitive purpose in the executive intelligence journey. These zones are designed to be consumed in sequence (Zone 1 through Zone 5) for the first-5-minute experience, while also supporting direct access to any zone for returning users who already have context.

Zone	Position	Description	Time Target
Zone 1: Attention Routing	Top of Command Center	AI-generated morning summary of top 3 items requiring immediate attention. Uses SynthesisEngine for natural language briefing grounded in today's signals.	Seconds 0-30
Zone 2: Portfolio Intelligence	Below summary	Visual grid of monitored accounts with movement indicators. Color-coded tiles by direction, sized by revenue impact. Answers "which accounts moved?"	Seconds 30-90
Zone 3: Narrative Feed	Center area	Ranked intelligence narratives with progressive disclosure. Sorted by composite score (confidence x priority x recency). Grouped by account for related signals.	Seconds 90-180
Zone 4: Action Queue	Right sidebar or panel	Prioritized, time-bound action recommendations extracted from all narratives. Shows urgency, owner (if assigned), and deadline. Persistent across screen transitions.	Seconds 180-240
Zone 5: Intelligence Health	Bottom strip or collapsible	System health indicators: signal freshness, evidence coverage, model confidence trends, user feedback stats. Transparent learning loop. Always visible but non-intrusive.	Seconds 240-300

5.2 Intelligence Hierarchy

The intelligence hierarchy defines how information is prioritized and surfaced within each zone. Rather than treating all intelligence equally, the hierarchy creates a clear priority order that reflects executive decision-making patterns. The hierarchy has four levels, each with distinct visual treatment and interaction patterns.

Level	Criteria	Visual Treatment	Disclosure Depth
L0: Critical Alert	Immediate action required; revenue at risk	Red accent, persistent notification, auto-expands in Zone 1	Full narrative + evidence + action at first view
L1: Priority Signal	Significant change detected; attention needed today	Amber accent, appears in morning summary, prominent in feed	Headline + confidence + top evidence + action
L2: Watch Item	Emerging pattern; monitor for escalation	Blue accent, appears in portfolio grid, standard in feed	Headline + confidence; evidence on expand
L3: Context Update	Informational; no immediate action required	Muted accent, appears in feed below fold, collapsible	Headline only; full narrative on explicit expand

5.3 Cards, Panels, and Feed Behavior

The card and panel system in Phase 1B must support three distinct interaction patterns: scanning (quick visual triage), reading (detailed understanding), and acting (executing a recommendation). The current IntelligenceNarrative component supports progressive disclosure but treats all narratives identically. Phase 1B introduces intelligence-level-dependent disclosure behavior.

L0 Cards (Critical Alerts) auto-expand to show full evidence and action at the first view. They cannot be dismissed without explicit acknowledgment and are visually distinct from all other intelligence through color, size, and animation. The card occupies approximately 40% more vertical space than L1 cards and includes a persistent "Action Required" badge. This ensures that critical intelligence is never accidentally overlooked during a quick scan.

L1 Cards (Priority Signals) show headline, confidence ring, top evidence source, and primary action at the default view. Evidence chain and detailed reasoning are available on click/expand. These cards form the bulk of the morning intelligence scan and should be designed for rapid reading with high information density per pixel.

L2 and L3 Cards show only headline and confidence at the default view, with full narrative available on explicit expansion. This creates a natural visual hierarchy where the VP Sales sees 3-5 L0/L1 cards prominently and 10-15 L2/L3 cards as a scannable list below.

Panel Behavior. The IntelligencePanel becomes the primary deep-dive surface. When a user clicks on any card (at any level), the panel slides in from the right with full narrative context. The panel automatically includes: the originating intelligence narrative, account context (industry, revenue band, current score), related signals (other signals for the same account), evidence chain with source links, and action recommendations with urgency indicators. The panel preserves its state when the user navigates to other screens, addressing the context loss problem identified in Chapter 3.

5.4 Navigation Reduction

The current 77+ screen navigation must be dramatically simplified for the executive experience. Phase 1B does not eliminate screens but restructures how they are accessed. The primary navigation reduction strategy has three components.

Component 1: Unified Command Center as Default Landing. The Command Center becomes the single default landing page for all users. Upon login, every user sees the Command Center first. This eliminates the need for users to choose between "Command Center," "AI Command Center," and "Dashboard" as their starting point. The Command Center's five zones (Attention, Portfolio, Feed, Actions, Health) replace the current multiple "home" screens.

Component 2: Contextual Navigation from Intelligence. When a user interacts with an intelligence narrative, the navigation surface changes to show only the screens relevant to that intelligence context. For example, clicking on a company intelligence narrative shows navigation options for: Company Detail, Account Intelligence, Contact Intelligence, Conversation Planner, and Email Generation. Screens unrelated to the current intelligence context are hidden, reducing cognitive load.

Component 3: Screen Consolidation. The following screen groups should be evaluated for consolidation in Phase 1B or Phase 2: "Command Center" + "AI Command Center" into a single enhanced Command Center; "Revenue Intelligence" + "Signal Intelligence" + "Account Intelligence" into a unified "Intelligence" view within the Command Center; "Companies" + "Company Detail" + "Account Intelligence" + "Account Ranking" into a unified company workspace; and "Intelligence Analytics" + "Intelligence Health" + "Intelligence Reasoning" into a consolidated "System Intelligence" panel.

Current Screens	Proposed Consolidation	Rationale
Command Center + AI Command Center	Enhanced Command Center	Single default landing with 5 zones
Revenue Intelligence + Signal Intelligence + Account Intelligence	Unified Intelligence View	Consolidated within Command Center zones
Companies + Company Detail + Account Intelligence + Account Ranking	Company Workspace	Single entity workspace with intelligence overlay

Current Screens	Proposed Consolidation	Rationale
Intelligence Analytics + Health + Reasoning	System Intelligence	Consolidated admin/ops panel
Contacts + Contact Detail + Contact Intelligence	Contact Workspace	Single entity workspace with relationship context

5.5 Context Preservation

Context preservation ensures that intelligence context is maintained across screen transitions. This is one of the most critical architectural changes in Phase 1B, as it directly addresses the "context loss" failure point identified in the 5-minute test. The implementation strategy uses three complementary mechanisms.

Mechanism 1: Intelligence Thread ID. Every intelligence narrative is assigned a unique thread ID that persists across screen transitions. When a user navigates from a Command Center narrative to a company detail screen, the thread ID is passed as a URL parameter and stored in the application state. The target screen reads this thread ID and automatically opens the IntelligencePanel with the originating narrative context. This creates a seamless "intelligence narrative to entity detail" flow without context loss.

Mechanism 2: Breadcrumb Intelligence Trail. A persistent breadcrumb trail at the top of every screen shows the intelligence path that led to the current view. For example: "Command Center > Meridian Technologies > CTO Departure Signal > Evidence Chain > SEC Filing." Each breadcrumb segment is clickable, allowing the user to navigate back to any point in the intelligence exploration without losing context.

Mechanism 3: Persistent Intelligence Panel. The IntelligencePanel, once opened, remains open during screen transitions. When the user navigates from Company A to Company B, the panel smoothly transitions to show Company B's intelligence while preserving the exploration depth (which disclosure level was expanded). This allows rapid comparison across accounts without repeatedly opening and closing the panel.

Chapter 6: Visual and Emotional Design

6.1 Why This Experience Feels Fundamentally Different

The fundamental question that Phase 1B must answer is: "Why would this experience feel fundamentally different from Salesforce, Gong, Clari, or traditional sales dashboards?" This is not a question about feature comparison or visual polish; it is a question about cognitive model and information flow

architecture. The answer lies in three structural differences that no amount of visual design can replicate.

Difference 1: Intelligence Push, Not Data Pull. Salesforce, Gong, and Clari are fundamentally data retrieval systems. The user asks a question ("show me my pipeline," "show me activity for this account") and the system returns data. DeepMindQ, after Phase 1B, must be an intelligence delivery system. The system tells the user what they need to know before they ask. The morning intelligence summary proactively surfaces critical information. The attention routing layer prioritizes items by executive impact, not data recency. The portfolio intelligence grid shows patterns the user did not explicitly search for. This shift from "user queries data" to "system delivers intelligence" is the foundational cognitive difference.

Difference 2: Reasoning Transparency, Not Black Box. Traditional AI-powered sales tools present AI insights as opaque recommendations: "AI suggests reaching out to this account." DeepMindQ makes the reasoning chain visible and auditable. The evidence chain shows exactly which signals, documents, and data points contributed to the recommendation. The confidence breakdown explains the four factors and their weights. The user can trace any intelligence item back to its source and evaluate the reasoning independently. This transparency creates trust that opaque AI recommendations cannot match.

Difference 3: Action Termination, Not Information Display. Every intelligence narrative in DeepMindQ ends with a specific, time-bound action. This is not a "suggested next step" or a "recommended action" that the user may or may not follow. It is a prioritized, urgency-rated, evidence-grounded directive that the AI has computed based on the current intelligence state. The action queue in Zone 4 aggregates all pending actions across the portfolio, creating an executable intelligence agenda. Traditional tools stop at "here is what is happening." DeepMindQ continues to "here is what you should do about it, and here is how urgent it is."

Dimension	Traditional Tools	DeepMindQ Phase 1B	Impact
Information Flow	User queries data	System delivers intelligence	Push vs. Pull cognitive model
AI Transparency	Opaque recommendations	Auditable reasoning chain	Evidence traceability
Action Model	User decides next steps	AI prioritizes actions by urgency	Recommendation-driven workflow
Learning Visibility	Hidden model updates	Transparent feedback loop	User sees how system improves
Navigation Model	Feature-based menu (77 screens)	Context-sensitive (5 zones)	Cognitive load reduction
Confidence Model	Score without explanation	Multi-factor with human-readable factors	Trust through transparency

6.2 Emotional Design Principles

The visual and emotional design of Phase 1B must evoke a specific feeling: "I am operating an intelligence command system, not using a software dashboard." This feeling is created through five emotional design principles that inform every visual and interaction decision.

Principle 1: Calm Authority. The interface should feel authoritative without being alarming. Critical intelligence is highlighted with purpose, not with noise. Colors are muted and intentional: red for critical, amber for priority, but these colors appear only where they matter. The overall palette remains calm (the cascade palette with warm, desaturated tones) to create a sense of controlled intelligence rather than chaotic alerts.

Principle 2: Intelligence Advantage. The experience should make the user feel like they have access to intelligence that their competitors do not. This is achieved through the evidence chain visibility, the "first to know" temporal indicators (showing when a signal was first detected relative to public knowledge), and the cross-account pattern detection that reveals insights invisible to single-account analysis tools.

Principle 3: Progressive Depth. The interface respects the user's time by providing exactly the right amount of information at each level. L0 critical items show everything. L3 context updates show almost nothing. The user controls the depth of exploration, and the system never forces information overload. This creates a sense of efficiency and control that is fundamentally different from the "wall of data" experience of traditional dashboards.

Principle 4: Trust Through Transparency. Every confidence score, evidence citation, and action recommendation is traceable to its source. The user never has to "trust the AI blindly" because the reasoning is always available. This principle is embedded in the progressive disclosure architecture: L2 shows the evidence, L3 shows the sources, L4 shows the raw data. Trust is earned through transparency, not assumed through branding.

Principle 5: Executive Efficiency. The entire experience is optimized for the executive time budget. The 5-minute journey is achievable because every zone, every card, and every interaction is designed to answer a specific question in minimal time. Navigation is reduced, context is preserved, and intelligence is prioritized. The VP Sales should feel that DeepMindQ respects their time more than any other tool in their stack.

6.3 Intelligence Advantage Feel

The "intelligence advantage" emotional goal is the differentiator that makes DeepMindQ feel like a category-defining product rather than a better CRM dashboard. This feeling is created through specific design patterns that make intelligence feel like a strategic asset rather than a feature.

First, the morning intelligence summary uses the SynthesisEngine to generate a natural language briefing that reads like an intelligence report from a human analyst, not a list of data points. The summary references specific signals, companies, and timeframes, making it feel grounded and authoritative.

Second, the portfolio intelligence grid reveals patterns that would be invisible in a traditional CRM: a cluster of hiring signals across an industry vertical, a correlation between news events and buying stage progression, or an anomaly in engagement patterns that suggests a competitor is making a move.

Third, the evidence chain with source citations creates a "knowledge edge" feeling: the user can see not just what the AI believes, but what evidence supports that belief, and access the original sources. Fourth, the learning transparency shows the user that their expertise is being amplified by the system: when they dismiss a false positive, the system acknowledges the feedback and adjusts. When they accept a recommendation, the system records the outcome. This creates a virtuous cycle where human expertise improves AI accuracy, and AI accuracy amplifies human expertise.

Chapter 7: Technical Implementation Strategy

7.1 Component Architecture

Phase 1B extends the Phase 1A component library with five new or enhanced components. These components are built on top of the existing intelligence OS design tokens and compose with the Phase 1A components (IntelligenceNarrative, EvidenceChain, ConfidenceIndicator, ActionCTA, IntelligencePanel). No existing components are removed or fundamentally restructured; instead, new components add the executive experience layer on top of the intelligence foundation.

Component	Status	Description	Data Dependencies
MorningBriefing	New	Zone 1 component. Generates and displays AI-synthesized morning intelligence summary. Uses SynthesisEngine with morning_brief type. Displays top 3 attention items with natural language reasoning.	IntelligenceNarrativeService (extended), SynthesisEngine
PortfolioGrid	New	Zone 2 component. Visual grid of monitored accounts as tiles. Each tile shows: company name, movement direction (color), top signal type, revenue band, intelligence score delta. Supports filtering and sorting.	Company signals, intelligence scores, revenue data
Intelligence Feed	Enhanced	Zone 3 component. Upgrades existing narrative list with: level-dependent disclosure (L0 auto-expand, L3 collapsed), composite ranking (confidence x priority x recency), account grouping for related signals, temporal badges ("new today", "escalated").	useIntelligenceNarratives (extended)

Component	Status	Description	Data Dependencies
ActionQueue	New	Zone 4 component. Extracts and aggregates all action recommendations from narratives. Shows: action description, urgency (time-bound), owning account, deadline, status (new/in-progress/completed). Persistent across screens.	ActionEngine, narrative actions
Intelligence HealthBar	Enhanced	Zone 5 component. Extends existing aggregate confidence with: signal freshness indicator, evidence coverage percentage, user feedback stats (accepted/dismissed ratio), model confidence trend (7-day sparkline).	Signal freshness, evidence coverage, feedback data

7.2 Data Flow Architecture

The Phase 1B data flow extends the Phase 1A pipeline with three new data paths while preserving the existing Signal-to-Component flow. The existing pipeline remains unchanged: CompanySignal -> GroundingEngine -> ConfidenceComputation -> ActionEngine -> IntelligenceNarrativeService -> /api/intelligence/narratives -> useIntelligenceNarratives -> CommandCenter. Phase 1B adds parallel data flows that feed the new components.

Flow	Input	Processing	Output	Zone
Morning Briefing	All active signals + account scores	SynthesisEngine (morning_brief type)	MorningBriefing component	Zone 1
Portfolio Grid	Company scores + signal counts + movement deltas	Aggregate query (company scores, recent signals)	PortfolioGrid component	Zone 2
Action Queue	All narrative actions (extracted during narrative generation)	IntelligenceNarrativeService (extended to emit action events)	ActionQueue component	Zone 4
Intelligence Health	Signal freshness + evidence coverage + feedback stats	Aggregate queries (signal dates, evidence counts, feedback table)	IntelligenceHealthBar component	Zone 5
Context Thread	Thread ID passed via URL params + app state	Client-side state management (zustand store)	IntelligencePanel + breadcrumb trail	Cross-screen

7.3 Intelligence Pipeline Extensions

Three extensions to the existing intelligence pipeline are required to support the Phase 1B components. These extensions are additive and do not modify the existing Phase 1A pipeline behavior. The existing narrative generation, confidence computation, and evidence collection mechanisms remain unchanged.

Extension 1: Morning Briefing Generation. A new SynthesisEngine brief type (morning_brief) that aggregates all active signals from the past 24 hours, cross-references them with account scores and movement data, and generates a natural language briefing. The briefing includes: total signal count by severity, top 3 attention items with reasoning, portfolio movement summary (positive/negative/neutral account counts), and actions approaching deadline. This is a new API endpoint: /api/intelligence/morning-brief.

Extension 2: Composite Narrative Ranking. The existing useIntelligenceNarratives hook returns narratives sorted by confidence. Phase 1B extends this with a composite ranking formula: $\text{rank} = \text{confidence} * \text{priorityWeight} * \text{recencyDecay}$, where priorityWeight maps critical=4, high=3, medium=2, low=1, and recencyDecay applies a 7-day half-life to ensure recent intelligence is ranked higher. This ranking is computed client-side using data already returned by the API.

Extension 3: Feedback Tracking API. A new API endpoint (/api/intelligence/feedback) that records user interactions with intelligence narratives: accept, dismiss, view, drill-down, and action-taken. This data feeds the IntelligenceHealthBar component (accepted/dismissed ratio) and is stored for future model fine-tuning. The existing intelligence feedback service is extended to support these new interaction types.

Chapter 8: Acceptance Criteria

Phase 1B implementation may not begin until this design document is reviewed and accepted. Upon acceptance, the following acceptance criteria must be met before Phase 1B can be considered complete. Each criterion is verifiable through either automated testing, user testing, or visual inspection.

8.1 Functional Acceptance Criteria

ID	Criterion	Description	Verification
A C- 1	Morning Briefing	Command Center displays an AI-generated morning summary at the top of Zone 1, synthesized from the past 24 hours of signals, with top 3 attention items identified by name and reasoning.	Manual inspection + API test
A C- 2	Portfolio Intelligence Grid	Zone 2 displays a visual grid of monitored accounts with movement direction indicators (positive/negative/neutral), color-coded tiles, and top signal type annotations for each account.	Manual inspection

ID	Criterion	Description	Verification
A C- 3	Level-Dependent Disclosure	Intelligence narratives display at different disclosure depths based on intelligence level: L0 auto-expands full narrative, L3 shows headline only. Level assignment is based on severity and recency.	Manual inspection + component test
A C- 4	Action Queue	Zone 4 displays all pending action recommendations across the portfolio with urgency indicators (time-bound), account association, and status tracking (new/in-progress/completed).	Manual inspection + API test
A C- 5	5-Minute Test	A VP Sales/CRO test participant can answer all 5 questions (attention, movement, reasoning, action, learning) within 5 minutes of first opening the application, measured by timed user testing.	User testing (3+ participants)
A C- 6	Context Preservation	Navigating from a Command Center narrative to a company detail screen preserves the intelligence thread context, shown via breadcrumb trail and auto-opened IntelligencePanel.	Manual inspection + E2E test
A C- 7	Feedback Loop	Accepting or dismissing an intelligence narrative records the feedback via the API, updates the IntelligenceHealthBar feedback stats, and provides user confirmation of the recording.	API test + manual inspection
A C- 8	Navigation Reduction	Default landing page is Command Center. Contextual navigation shows only relevant screens when an intelligence narrative is active. Screen consolidation plan is documented for Phase 2.	Manual inspection

8.2 Technical Acceptance Criteria

ID	Criterion	Description	Verification
T C- 1	Pipeline Integrity	All 5 zones consume real intelligence data from the pipeline. Zero static/template data in any zone.	Code audit
T C- 2	Test Coverage	New components have test coverage: MorningBriefing (5+ tests), PortfolioGrid (5+ tests), ActionQueue (5+ tests). Total new tests ≥ 15 .	Test execution
T C- 3	TypeScript	Zero TypeScript compilation errors after Phase 1B implementation.	<code>npx tsc --noEmit</code>
T C- 4	Build	Production build succeeds with zero errors.	<code>npx next build</code>

ID	Criterion	Description	Verification
T C- 5	Lint	Zero lint errors.	bun run lint
T C- 6	Performance	Command Center initial render completes within 3 seconds on standard hardware (4-core, 8GB). Intelligence narratives load within 5 seconds.	Performance measurement
T C- 7	Regression	All existing Phase 1A tests pass (1888+ tests, 0 new failures).	Test execution
T C- 8	API Endpoints	New API endpoints (/api/intelligence/morning-brief, extended /api/intelligence/feedback) are authenticated, have error handling, and return correct data shapes.	API test

8.3 Known Limitations for Phase 1B

The following limitations are acknowledged for Phase 1B and will be addressed in subsequent phases. They should not block Phase 1B closure but must be documented for planning purposes.

Limitation	Description	Target Phase
Real-Time Updates	WebSocket-based real-time intelligence updates are not included in Phase 1B. The system continues to use polling-based refresh (manual or timed).	Phase 2
Full Screen Consolidation	Phase 1B documents the consolidation plan and implements the Command Center as default landing. Full screen elimination/merger (77 screens to ~40) is a Phase 2 scope.	Phase 2
Cross-Account Intelligence	Cross-account pattern detection and industry-vertical analysis are displayed at a basic level. Advanced cross-account correlation (multi-account buying signals, competitive displacement patterns) is Phase 2.	Phase 2
Mobile Optimization	The Phase 1B Command Center is designed for desktop executive use. Mobile-responsive optimization is deferred to Phase 3.	Phase 3
Multi-User Collaboration	Intelligence sharing, team-based action assignment, and collaborative intelligence review are not included in Phase 1B.	Phase 3